

# CLASS -10 (2025-26)

## INPUT IN JAVA

### CHAPTER 5

### Assignments:-

#### A. Tick (✓) the correct answer

**Question 1.** Which of the following types of error will the following code give?

```
for(int i=1;1>10;i++)
```

- a. Syntax
- b. Runtime
- c. Logical
- d. None of these

**Answer:-** c. Logical

---

**Question 2.** The statement `Scanner sc = new Scanner(System.in);` will give a/an \_\_\_\_ error.

- a. Syntax
- b. Runtime
- c. Logical
- d. None of these

**Answer:-** a. Syntax

#### **Explanation:**

The class name `Scanner` is case-sensitive in Java. Writing `scanner` with a lowercase 's' causes a **syntax error** because Java doesn't recognize it as a valid class name. The correct syntax should be:

```
Scanner sc = new Scanner(System.in);
```

---

**Question 3.** ..... are non-executable statements used to explain the code.

- a. Comments
- b. Errors
- c. Logical
- d. None

**Answer:-** a. Comments

---

**Question 4.** Which of the following is not used to write a comment?

- a. `/**`
- b. `//`
- c. `/`
- d. `/*`

**Answer:-** c. `/`

---

**Question 5.** Which of the following is used to input a float value using `Scanner` class?

- a. `next()`
- b. `nextInt()`
- c. `nextDouble()`
- d. `nextFloat()`

**Answer:-** d. `nextFloat()`

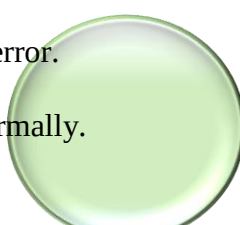
### **B. Fill in the blanks**

**Question 1.** Forgetting to put a semicolon at the end of a statement is a type ..... error.

**Answer:-** Syntax

**Question 2.** An ..... is a situation in which the execution of a program stops abnormally.

**Answer:-** Exception



**Question 3.** ..... errors are also called semantic errors.

**Answer:-** Logical

**Question 4.** ..... comments begin with /\* and end with \*/.

**Answer:-** Multiline

**Question 5.** We use ..... arguments to input values in String format in the main() method.

**Answer:-** String[] args

---

## C. Short Answer Type Questions

**Question 1.** What is the difference between multiline comments and documentation comments?

**Answer:-** Multiline comments (/\*...\*/) are used to comment out multiple lines, while documentation comments (/\*\*...\*/) are used to generate official documentation using javadoc.

**Question 2.** Write the syntax to input a Short type value using the Scanner class.

**Answer:-**

```
Scanner sc = new Scanner(System.in);  
short value = sc.nextShort();
```

**Question 3.** What are the three types of errors?

**Answer:-** Syntax error, Runtime error, and Logical error.

**Question 4.** Define logical error. Give an example.

**Answer:-** A logical error occurs when the program compiles and runs but gives the wrong output.

**Example:** Calculating average as `total / number + 2` instead of `(total + 2) / number`.

**Question 5.** Write the difference between try and catch.

**Answer:-** The try block contains code that might throw an exception. The catch block handles the exception.

---

## D. Assertion and Reasoning Based Question

**Question 1.**

**Assertion (A):** A package is a collection of Java classes and other packages having similar functionality.

**Reason (R):** Each class within a package cannot use the public properties and behaviours of other classes.

- a. Both A and R are true, and R is the correct explanation of A.
- b. Both A and R are true, but R is not the correct explanation of A.
- c. A is true, but R is false.
- d. A is false, but R is true.

**Answer:-** c. A is true, but R is false.

\*\*\*\*\*

---

## Some extra Questions and Answers

**Question 1:** What will happen if we write:

```
scanner sc = new Scanner(System.in);
```

**Options:**

- a. Runtime error
- b. Logical error
- c. Syntax error
- d. No error

**Answer:-** c. Syntax error



**Explanation:** Java is case-sensitive. `scanner` with a small 's' is not a valid class name. The correct class name is `Scanner`.

### Question 2:

Which of the following will **definitely** cause a **runtime error** if executed without user input?

```
Scanner sc = new Scanner(System.in);  
int x = sc.nextInt();
```

#### Options:

- a. If user types a number
- b. If user types a float value
- c. If user types a string
- d. If user types whitespace

**Answer:-** c. If user types a string

**Explanation:** `nextInt()` expects an integer. Typing a string causes `InputMismatchException` at runtime.

### Question 3:

What type of error is caused by the code below?

```
int x = 10;  
System.out.println("Result: " + x / 0);
```

#### Options:

- a. Syntax error
- b. Runtime error
- c. Logical error
- d. No error

**Answer:-** b. Runtime error

**Explanation:** Division by zero leads to `ArithmeticException` during program execution.

### Question 4:

Which of the following is **not** a valid way to write a comment in Java?

#### Options:

- a. `// This is a comment`
- b. `/* This is a comment */`
- c. `/* This is a documentation comment */`
- d. `/ This is a comment /`

**Answer:-** d. `/ This is a comment /`

**Explanation:** This is not a valid syntax. A single slash `/` is not recognized as a comment indicator.

### Question 5:

What does the following code print if the user enters 5 and 3?

```
Scanner sc = new Scanner(System.in);  
int a = sc.nextInt();  
int b = sc.nextInt();  
System.out.println((a > b) ? a : b);
```

#### Options:

- a. 3
- b. 5
- c. a
- d. b

**Answer:-** b. 5

**Explanation:** The ternary operator returns the greater of the two values.

## Primitive Data Types – Keyboard Input Questions



### Question 1:

Which method is used to input an integer value in Java using the Scanner class?

**Options:**

- a. next()
- b. nextInt()
- c. nextInteger()
- d. readInt()

**Answer:-** b. nextInt()

---

### Question 2:

Which Scanner method is used to input a float value?

**Options:**

- a. nextFloat()
- b. nextDouble()
- c. nextReal()
- d. nextFlt()

**Answer:-** a. nextFloat()

---

### Question 3:

Which method is used to input a boolean value (true/false)?

**Options:**

- a. next()
- b. nextBoolean()
- c. nextBool()
- d. readBoolean()

**Answer:-** b. nextBoolean()

---

### Question 4:

Which method of Scanner is used to input a byte value?

**Options:**

- a. nextByte()
- b. nextSmall()
- c. byte()
- d. getByte()

**Answer:-** a. nextByte()

---

### Question 5:

Which Scanner method allows input of a single word (string without space)?

**Options:**

- a. nextString()
- b. next()
- c. nextLine()
- d. get()

**Answer:-** b. next()

---

### Question 6:

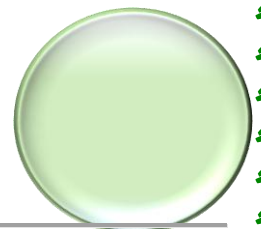
Which Scanner method is used to input an entire line including spaces?

**Options:**

- a. nextWord()
- b. readLine()
- c. next()
- d. nextLine()

**Answer:-** d. nextLine()

---



### Question 7:

How do you input a `short` value using Scanner?

**Options:**

- a. `nextInt()`
- b. `nextByte()`
- c. `nextShort()`
- d. `nextSmall()`

**Answer:-** c. `nextShort()`

---

### Question 8:

Which Scanner method is used to input a double value?

**Options:**

- a. `nextFloat()`
- b. `nextDouble()`
- c. `nextReal()`
- d. `nextDecimal()`

**Answer:-** b. `nextDouble()`

---

### Question 9:

Which primitive data type does `nextLong()` help to input?

**Options:**

- a. `float`
- b. `long`
- c. `double`
- d. `int`

**Answer:-** b. `long`

---

### Question 10:

Which method would best input a character using Scanner?

**Options:**

- a. `nextChar()`
- b. `next().charAt(0)`
- c. `nextCharacter()`
- d. `readChar()`

**Answer:-** b. `next().charAt(0)`

---

## Non-Primitive Data Types – Keyboard Input Questions

---

### Question 11:

How can you input a `String` in Java using Scanner that includes spaces?

**Options:**

- a. `next()`
- b. `readString()`
- c. `nextLine()`
- d. `scanString()`

**Answer:-** c. `nextLine()`

---

### Question 12:

Which of the following is **not** a primitive data type?

**Options:**

- a. `int`
- b. `String`
- c. `float`



d. boolean

**Answer:-** b. String

---

**Question 13:**

What is the data type of `Scanner sc = new Scanner(System.in);`?

**Options:**

- a. Primitive
- b. Non-primitive
- c. Keyword
- d. Operator

**Answer:-** b. Non-primitive

---

**Question 14:**

Which Java package must be imported to use the Scanner class?

**Options:**

- a. java.input.Scanner
- b. java.io
- c. java.util
- d. java.lang

**Answer:-** c. java.util

---

## Short answer questions with answers

**Question 1.**

Which class is commonly used in Java to take keyboard input from the user?

**Answer:-** Scanner class.

---

**Question 2.**

Which package must be imported to use the Scanner class in Java?

**Answer:-** java.util package.

---

**Question 3.**

Write the syntax to create a Scanner object.

**Answer:-** `Scanner sc = new Scanner(System.in);`

---

**Question 4.**

How do you input an integer value using Scanner?

**Answer:-** `int num = sc.nextInt();`

---

**Question 5.**

How can you input a string that includes spaces using the Scanner class?

**Answer:-** `String text = sc.nextLine();`

---

**Question 6.**

Write the code to input a float value using Scanner.

**Answer:-** `float f = sc.nextFloat();`

---

**Question 7.**

How can you input a single character using Scanner?

**Answer:-** `char ch = sc.next().charAt(0);`

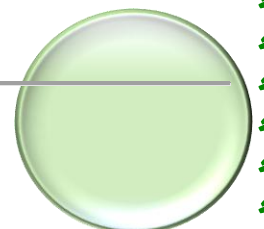
---

**Question 8.**

What is the Scanner method to read a boolean value?

**Answer:-** `nextBoolean()`

---



**Question 9.**

What is the purpose of `System.in` in the Scanner object?

**Answer:-** It represents the standard input stream (keyboard input).

**Question 10.**

Can the Scanner class be used to input non-primitive data types like String?

**Answer:-** Yes, using `next()` or `nextLine()` methods.

\*\*\*\*\*

